

Habib Shakour

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Education

University of Michigan - Ann Arbor

Bachelor of Science (BS in Computer Science)

Ann Arbor, MI

August 2026

- GPA: 3.0/4.0
- Course Highlights: Data Structures and Algorithms, Operating Systems, Intro to AI, Computer Security, Foundations of Computer Science, Database Management Systems, Web Systems, Computer Networking

Professional Experience

Cadence Design Systems

Software Engineering Intern - System Verification Group (SVG)

San Jose, CA

May 2025 – Aug 2025

- Designed an AI tool to configure verification IP parameters to customer devices with greater speed and accuracy
- Presented Autoconfig AI project results to Cadence executives, demonstrating business and technical value
- Collaborated globally with Indian + Brazilian teams to coordinate project responsibilities for maximum productivity
- Gained hands-on experience with EDA workflows, leveraging tools for circuit design, verification, and optimization

University of Michigan Department of Mechanical Engineering

Research Assistant

Ann Arbor, MI

June 2023 – Nov 2025

- Optimized data measuring for a machine learning algorithm through investigating fNIRS Technology
- Developed C++ code in Unreal Engine to simulate disaster relief with machine learning using VIVE VR headsets
- Demonstrated Unreal Engine simulation to PCAST, addressing questions to clarify technical aspects
- Trained incoming students on lab development tools and workflows to accelerate onboarding

Organizations

Michigan Autonomous Aerial Vehicles

Software Team Lead

Ann Arbor

Sept. 2023 – May 2026

- Directed a team of 4 students to create a functioning, autonomous navigation flight stack to fly any given course
- Co-developed a Computer Vision algorithm that identifies airdrop locations using Python with Open CV
- Built a functional simulation of SUAS course with drone functionality for testing C++ flight algorithms through ROS
- Implemented a comprehensive plan to maximize the productivity of 10 team members over a 12-week period

Skills

- **Programming Languages:** C++, C, Python, Java, R, MATLAB, JavaScript, SQL, HTML, CSS, System Verilog
- **Software/Applications:** Robot Operating System (ROS), LaTeX, OpenCV2, TensorFlow, PyTorch, Unreal Engine 4, Git, REACT, REST, SQL*Plus, MongoDB, AWS, gdb, Linux, UVM, Visual Studio Code, Jira

Projects

Glucose Tracker Web Application Prototype

Apr. 2026

- Developed a website to assist diabetics using client-side dynamic pages and a REST API with python back end
- Managed a relational database of users' biometric info, accounts, & meals using SQLite to ensure data integrity
- Coordinated development with 3 other students, contributing to timely milestones and efficient project execution.
- Ensured project integrity through end-to-end testing with Cypress and deployed project to AWS for scalability

Automatic Drone Communication Network for Minefield Mapping

Feb. 2026

- Led a team of 9 students in designing a TCP network for a fleet of four autonomous exploration drones.
- Implemented a Manager-Worker architecture to handle job distribution, fault tolerance, and concurrency
- Designed algorithm for segmenting minefield into grid and mapping all detected mines with Python

Article Search Engine

Dec. 2024

- Implemented a segmented inverted index of over 3000 articles using a hadoop MapReduce pipeline
- Ran multiple Index servers with Search server concurrently using multithreading for efficient lookup
- Built a user-friendly web interface using server-side dynamic pages for easy search and display of results